



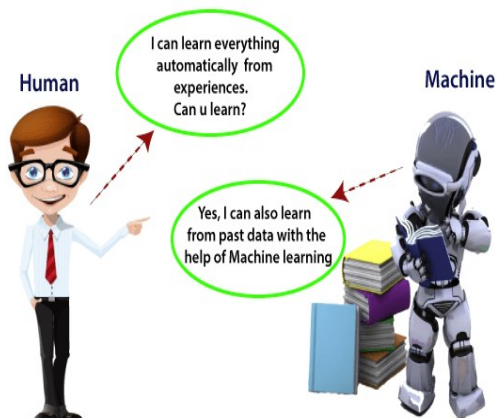
ICE-3201

**Artificial Intelligence and
Neural Networks****Lecture-3****Machine Learning****Course Teacher****Prof. Dr. Md. Ashikur Rahman Khan****What is Machine Learning**

In the real world, we are surrounded by **humans who can learn everything from their experiences with their learning capability**, and we have computers or machines which work on our instructions.

But can a machine also learn from experiences or past data like a human does?

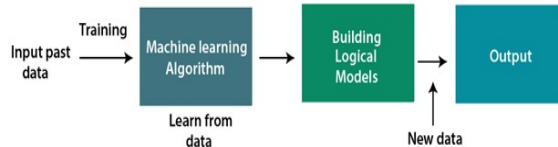
So here comes the role of **Machine Learning**.



How does Machine Learning work

A ML system builds prediction models, learns from previous data, and predicts the output of new data whenever it receives it.

The amount of data helps to build a better model that accurately predicts the output, which in turn affects the accuracy of the predicted output.



Features of Machine Learning:

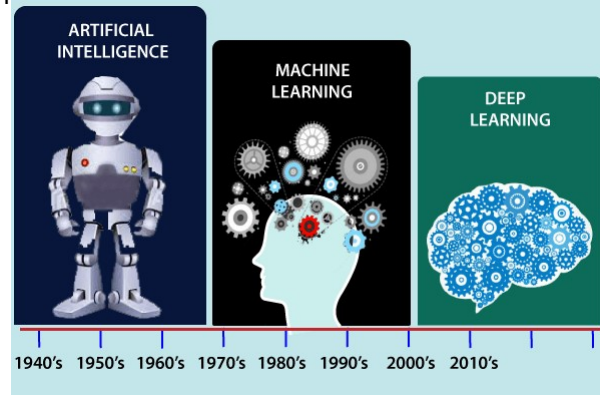
Features of Machine Learning:

- Machine learning uses data to detect various patterns in a given dataset.
- It can learn from past data and improve automatically.
- It is a data-driven technology.
- Machine learning is much similar to data mining as it also deals with the huge amount of the data.

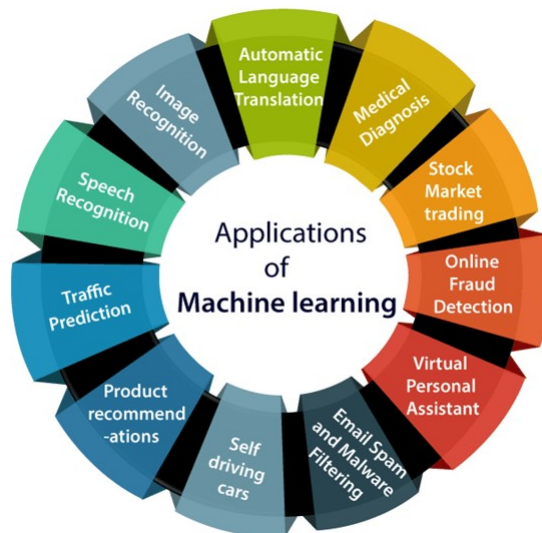
History of Machine Learning

Before some years (about 40-50 years), machine learning was science fiction, but today it is the part of our daily life. Machine learning is making our day to day life easy from **self-driving cars** to **Amazon virtual assistant "Alexa"**.

Below some milestones are given which have occurred in the history of machine learning:

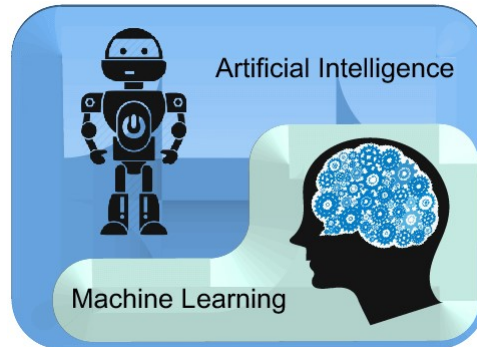


Applications of Machine Learning



Key differences between Artificial Intelligence (AI) and Machine learning (ML):

- *AI is a bigger concept to create intelligent machines that can simulate human thinking capability and behavior, whereas, machine learning is an application or subset of AI that allows machines to learn from data without being programmed explicitly.*
- *Machine learning is a subfield of artificial intelligence, which enables machines to learn from past data or experiences without being explicitly programmed.*



Key differences between Artificial Intelligence (AI) and Machine learning (ML):

AI is a technology which enables a machine to simulate human behavior.	ML is a subset of AI which allows a machine to automatically learn from past data without programming explicitly.
The goal of AI is to make a smart computer system like humans to solve complex problems.	The goal of ML is to allow machines to learn from data so that they can give accurate output.
In AI, we make intelligent systems to perform any task like a human.	In ML, we teach machines with data to perform a particular task and give an accurate result.
Machine learning and deep learning are the two main subsets of AI.	Deep learning is a main subset of machine learning.
AI has a very wide range of scope.	Machine learning has a limited scope.
AI is working to create an intelligent system which can perform various complex tasks.	Machine learning is working to create machines that can perform only those specific tasks for which they are trained.
AI system is concerned about maximizing the chances of success.	Machine learning is mainly concerned about accuracy and patterns.

Key differences between Artificial Intelligence (AI) and Machine learning (ML):

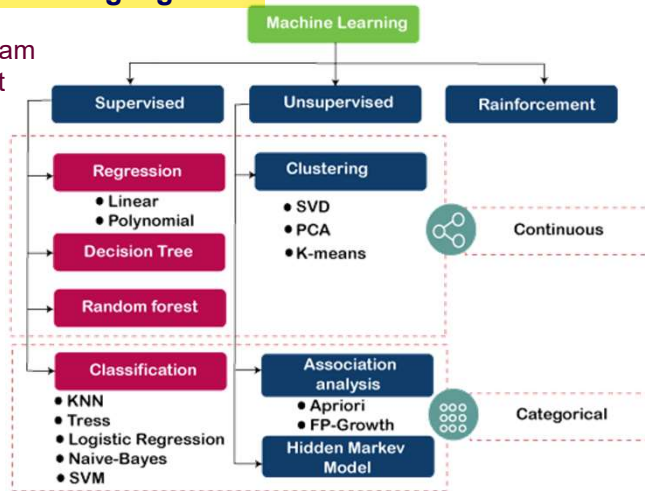
The main applications of AI are Siri, customer support using chatbots, Expert System , Online game playing, intelligent humanoid robot, etc.	The main applications of machine learning are Online recommender system, Google search algorithms, Facebook auto friend tagging suggestions , etc.
It includes learning, reasoning, and self-correction.	It includes learning and self-correction when introduced with new data.
AI completely deals with Structured, semi-structured, and unstructured data.	Machine learning deals with Structured and semi-structured data.

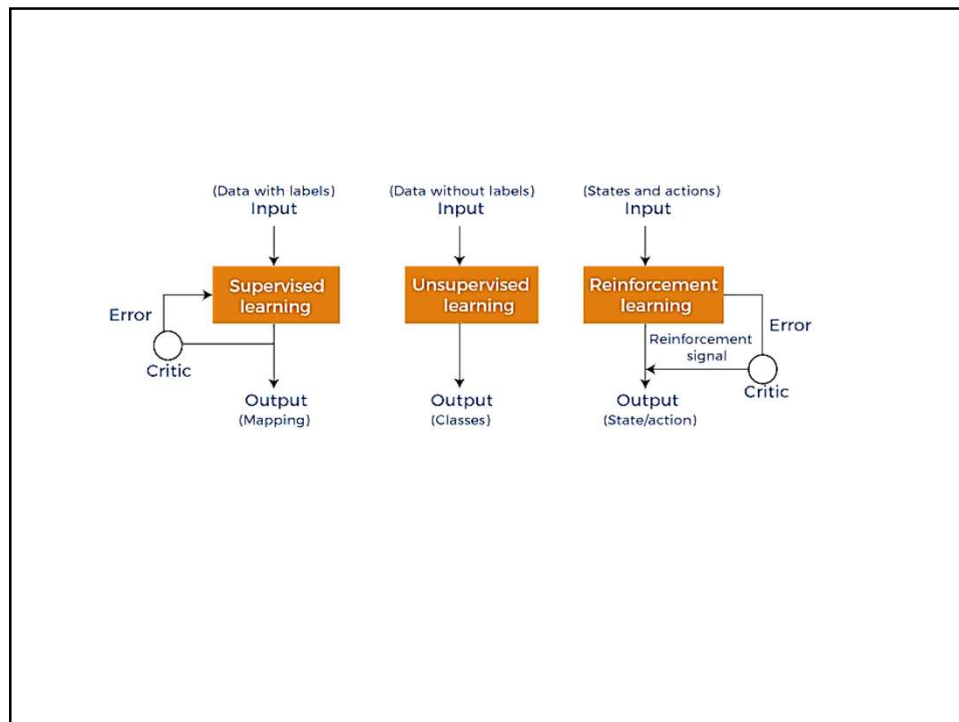
Types of Machine Learning Algorithms

Machine Learning Algorithm can be broadly classified into three types:

- 1. Supervised Learning Algorithms**
- 2. Unsupervised Learning Algorithms**
- 3. Reinforcement Learning algorithm**

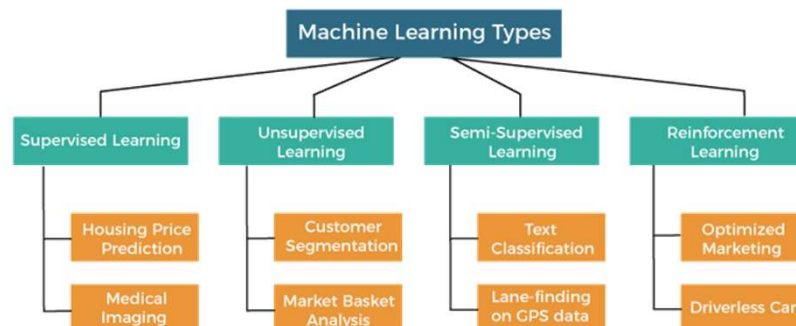
The below diagram illustrates the different ML algorithm, along with the categories:





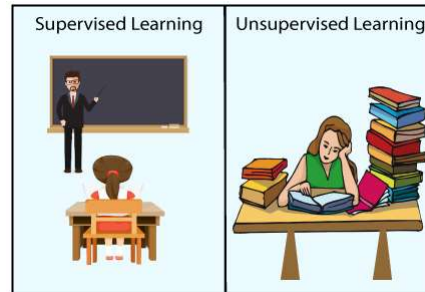
Based on the methods and way of learning, machine learning is divided into mainly four types, which are:

1. Supervised Machine Learning
2. Unsupervised Machine Learning
3. Semi-Supervised Machine Learning
4. Reinforcement Learning



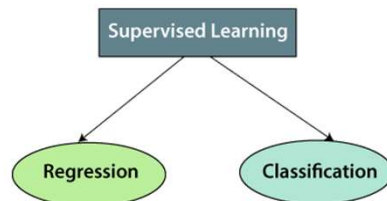
Types of Machine Learning Algorithms

Difference between Supervised and Unsupervised Learning

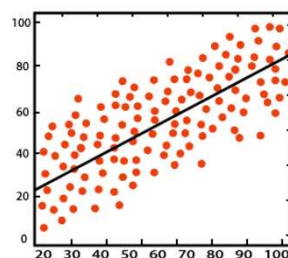


Types of supervised Machine learning Algorithms

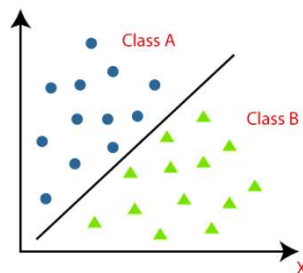
Supervised learning can be further divided into two types of problems:



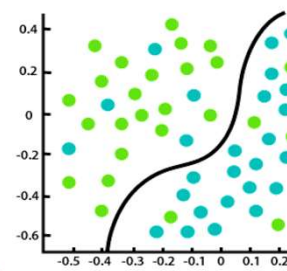
1. Classification and
2. Regression



Regression



Classification



Classification

Types of supervised Machine learning Algorithms

1. Regression

Regression is a process of finding the correlations between dependent and independent variables. Regression algorithms are used if there is a relationship between the input variable and the output variable. The task of the Regression algorithm is to find the mapping function to map the input variable(x) to the continuous output variable(y).

It helps in predicting the continuous variables such as prediction of **Market Trends**, prediction of House prices, etc.

It is used to predict the continuous values such as price, salary, age, weight, height, Weather forecasting, Market Trends, etc.

Examples:

- Prediction price of house or property.
- Prediction USD/EUR rate for tomorrow
- Predicting stock market price
- Predicting whether a patient has a cancer.

Regression is further classified into:

(i) Linear Regression

(ii) Logistics Regression

Types of supervised Machine learning Algorithms

2. Classification

Algorithm is trained to classify something into some class or category.

Classification algorithms are used to solve the classification problems in which the output variable is categorical, such as "Yes" or No, Male or Female, Red or Blue, etc. The classification algorithms predict the categories present in the dataset.

Such as Spam Detection, Email filtering, etc.

Examples:

- Classifying whether an email is spam or not.
- Classifying whether a patient has disease or not
- Identify & classify different objects in an image like human, animal, car etc.
- Classify photos of animals into classes eg. dogs and cats.
- New product recommendation on ecommerce websites based on your previous purchase and browsing history

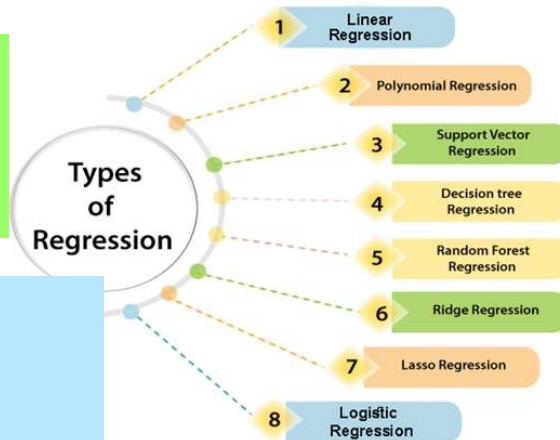
Types of Regression

There are various types of regressions which are used in data science and machine learning.

Here we are discussing some important types of regression which are given below:

- (a) Simple Linear Regression
- (b) Multiple Linear Regression
- (c) Non-Linear Regression
- (d) Polynomial Regression
- (e) Logistic Regression

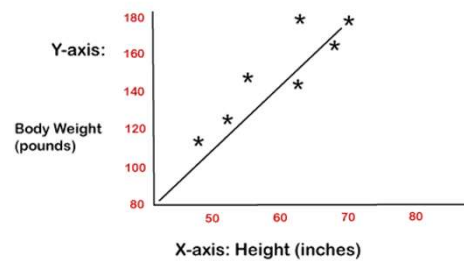
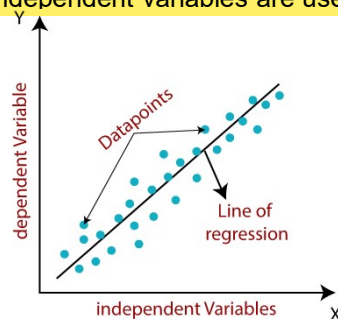
- i. Support Vector Regression
- ii. Decision Tree Regression or Regression Tree
- iii. Random Forest Regression
- iv. Ridge Regression
- v. Lasso Regression:
- vi. Bayesian Linear Regression



Linear regression is further divided into two types:

•**Simple Linear Regression:** In simple linear regression, a single independent variable is used to predict the value of the dependent variable.

•**Multiple Linear Regression:** In multiple linear regression, more than one independent variables are used to predict the value of the dependent variable.



Simple Linear Regression, $y = a_0 + a_1x + \epsilon$

Multiple Linear Regression,

$$y = a_0 + a_1 * x_1 + a_2 * x_2 + \dots + a_n * x_n$$

Linear regression is further divided into two types:

Mathematically, we can represent a linear regression as:

$$y = a_0 + a_1x + \epsilon$$

Here,

Y= Dependent Variable (Target Variable)

X= Independent Variable (predictor Variable)

a_0 = intercept of the line (Gives an additional degree of freedom)

a_1 = Linear regression coefficient (scale factor to each input value).

ϵ = random error

The values for x and y variables are training datasets for Linear Regression model representation.

Linear Regression Line

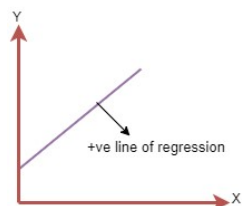
A linear line showing the relationship between the dependent and independent variables is called a **regression line**. A regression line can show two types of relationship:

•Positive Linear Relationship:

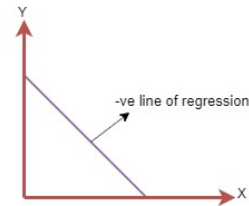
If the dependent variable increases on the Y-axis and independent variable increases on X-axis, then such a relationship is termed as a Positive linear relationship.

•Negative Linear Relationship:

If the dependent variable decreases on the Y-axis and independent variable increases on the X-axis, then



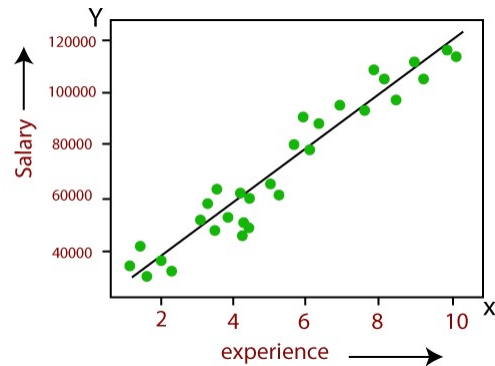
The line equation will be: $Y = a_0 + a_1x$



The line of equation will be: $Y = -a_0 + a_1x$

Example: Suppose there is a marketing company A, who does various advertisement every year and get sales on that. The below list shows the advertisement made by the company in the last 5 years and the corresponding sales:

Advertisement	Sales
\$90	\$1000
\$120	\$1300
\$150	\$1800
\$100	\$1200
\$130	\$1380
\$200	??



• Below is the mathematical equation for Linear regression:

$$Y = ax + b$$

Here, Y = dependent variables (target variables),

X = Independent variables (predictor variables),

a and b are the linear coefficients

Transformation of Non-Linear Regression to Linear Regression

Transformations

- The trick is to convert non-linear data to linear data that can be handled using the linear regression method.
- Let us consider an exponential function $y = ae^{bx}$.
- The transformation can be done by applying log function to both sides to get:

$$\ln(y) = \ln(a) + \ln(e^{bx})$$

$$\ln(y) = \ln(a) + bx * \ln(e)$$

$$\ln(y) = \ln(a) + bx$$

Since $\ln(e) = 1$

Activate Windows
Go to Settings to activate Windows.

Transformation of Non-Linear Regression to Linear Regression

Regression on Transformed Variables

Nonlinearity (in parameters) is visually determined from the scatter diagram, and sometimes, because of prior experience or underlying theory, we know in advance that the model is nonlinear

Nonlinear	Transformation	Linear
$Y = \beta_0 \beta_1^X \epsilon$	$\text{Log}(Y) = \log(\beta_0) + X \log(\beta_1) + \log(\epsilon)$	$Y^* = \beta_0^* + \beta_1^* X + \epsilon^*$
$Y = \beta_0 X^{\beta_1} \epsilon$	$\log(Y) = \log(\beta_0) + \beta_1 \log(X) + \log(\epsilon)$	$Y^* = \beta_0^* + \beta_1^* X^* + \epsilon^*$
$Y = \beta_0 e^{\beta_1 X} \epsilon$	$\ln(Y) = \ln(\beta_0) + \beta_1 X + \ln(\epsilon)$	$Y^* = \beta_0^* + \beta_1^* X + \epsilon^*$
$Y = \frac{1}{\beta_0 + \beta_1 X + \epsilon}$	$\frac{1}{Y} = \beta_0 + \beta_1 X + \epsilon$	$Y^* = \beta_0 + \beta_1 X + \epsilon$
$Y = \frac{X}{\beta_0 X + \beta_1 + X \epsilon}$	$\frac{1}{Y} = \beta_0 + \beta_1 \frac{1}{X} + \epsilon$	$Y^* = \beta_0 + \beta_1 X^* + \epsilon$

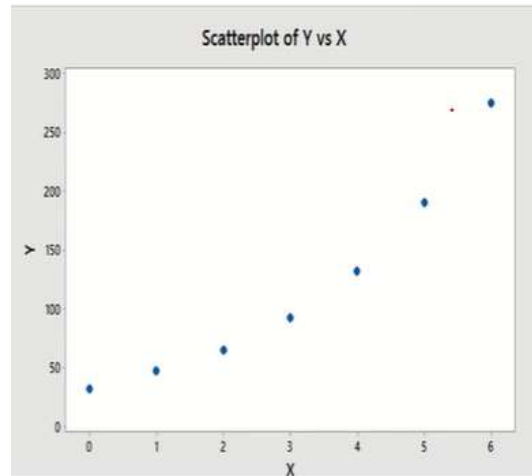
2. Non-Linear Regression to Linear Regression Transformation

EXAMPLE:- The number (Y) of bacteria per unit volume present in a culture after X hours is given in the following table. Fit a least square curve having the form $Y = \beta_0 \beta_1^X \epsilon$ to the data. Estimate the value of Y when X=7.

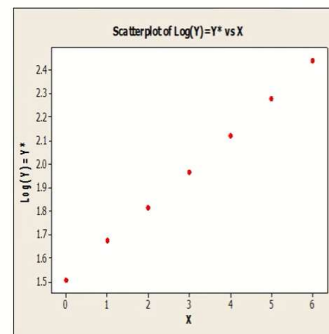
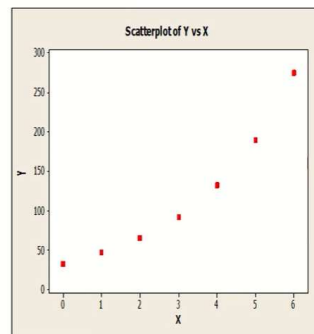
Y	X
32	0
47	1
65	2
92	3
132	4
190	5
275	6

2. Non-Linear Regression to Linear Regression Transformation (Contd.)

Y	X
32	0
47	1
65	2
92	3
132	4
190	5
275	6

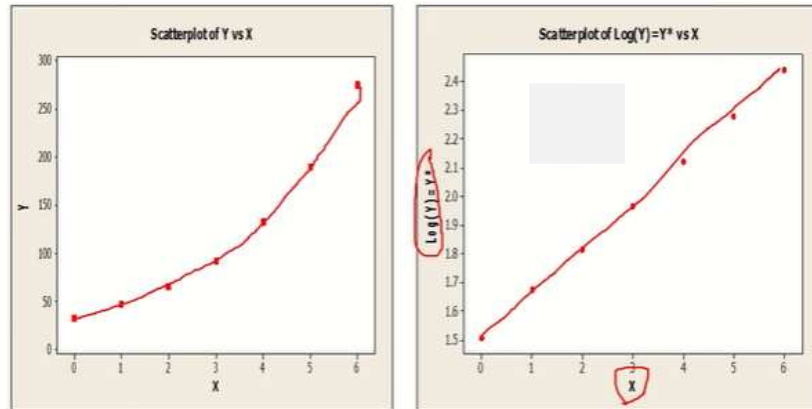


2. Non-Linear Regression to Linear Regression Transformation (Contd.)



$Y = \beta_0 \beta_1^X \epsilon$	$\text{Log}(Y) = \log(\beta_0) + X \log(\beta_1) + \log(\epsilon)$	$Y^* = \beta_0^* + \beta_1^* X + \epsilon^*$
----------------------------------	--	--

2. Non-Linear Regression to Linear Regression Transformation (Contd.)



$Y = \beta_0 \beta_1^X \epsilon$	$\log(Y) = \log(\beta_0) + X \log(\beta_1) + \log(\epsilon)$	$Y^* = \beta_0^* + \beta_1^* X + \epsilon^*$
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2. Non-Linear Regression to Linear Regression Transformation (Contd.)

$Y = \beta_0 \beta_1^X \epsilon$	$\log(Y) = \log(\beta_0) + X \log(\beta_1) + \log(\epsilon)$	$Y^* = \beta_0^* + \beta_1^* X + \epsilon^*$
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X	Y	Y* = log(Y)
0	32	1.50515
1	47	1.6721
2	65	1.81291
3	92	1.96379
4	132	2.12057
5	190	2.27875
6	275	2.43933

Estimation of regression of **X on Y*** by least Square

$$Y = A + Bx$$

In this equation, if we know the value of many Y and many x, can we find the value of A and B?

Similarly, for the equation $Y^* = \beta_0^* + \beta_1^* X$
we can determine β_0^* and β_1^*

2. Non-Linear Regression to Linear Regression Transformation (Contd.)

$Y = \beta_0 \beta_1^X \epsilon$	$\text{Log}(Y) = \log(\beta_0) + X \log(\beta_1) + \log(\epsilon)$	$Y^* = \beta_0^* + \beta_1^* X + \epsilon^*$
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$$Y = \beta_0 \beta_1^X \epsilon$$

$$\beta_0 = \log(\beta_0^*) = \log(1.51) = 32.36$$

$$\beta_1 = \log(\beta_1^*) = \log(0.154) = 1.43$$

$$Y = \beta_0 \beta_1^X \epsilon = 32.36(1.43)^X$$

Hence, the estimated growth after 7 hours

$$Y = 32.36(1.43)^X$$

$$Y = 32.36(1.43)^7$$

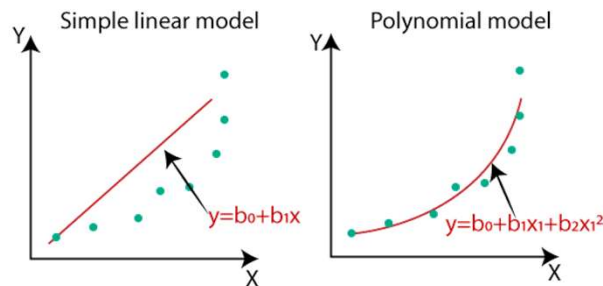
$$Y = 395.70 \text{ Ans.}$$

3. Polynomial Regression

Polynomial Regression is a regression algorithm that models the relationship between a dependent (y) and independent variable (x) as nth degree polynomial.

The Polynomial Regression equation is given below:

$$y = b_0 + b_1 x_1 + b_2 x_1^2 + b_3 x_1^3 + \dots + b_n x_1^n$$



Equation of the Polynomial Regression Model:

Simple Linear Regression equation: $y = b_0 + b_1 x$ (a)

Multiple Linear Regression equation: $y = b_0 + b_1 x + b_2 x_2 + b_3 x_3 + \dots + b_n x_n$ (b)

Polynomial Regression equation: $y = b_0 + b_1 x + b_2 x^2 + b_3 x^3 + \dots + b_n x^n$ (c)

3. Polynomial Regression

Polynomial Regression

- It can handle non-linear relationships among variables by using n^{th} degree of a polynomial.
- Instead of applying transformations, polynomial regression can be directly used to deal with different levels of curvilinearity.
- For example, the second-degree polynomial (called quadratic transformation) is given as:

$$y = a_0 + a_1x + a_2x^2$$
and third degree polynomial is called cubic transformation given as:

$$y = a_0 + a_1x + a_2x^2 + a_3x^3$$
- Generally, polynomials of maximum degree 4 are used, as higher order polynomials take some strange shapes and make the curve more flexible.
- It leads to a situation of overfitting and hence is avoided.

4. Logistic Regression

- Logistic regression algorithm works with the categorical variable such as 0 or 1, Yes or No, True or False, Spam or not spam, etc.
- It is a predictive analysis algorithm which works on the concept of probability.
- Logistic regression is used when the dependent variable is binary (0/1, True/False, Yes/No) in nature.

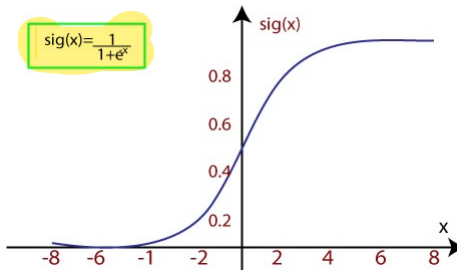
There are three types of logistic regression:

- Binary(0/1, pass/fail)
- Multi(cats, dogs, lions)
- Ordinal(low, medium, high)

4. Logistic Regression

- Logistic regression uses **sigmoid function** or **logistic function** which is a complex cost function. This sigmoid function is used to model the data in logistic regression. The function can be represented as:

$$f(x) = \frac{1}{1 + e^{-x}}$$



Where,

$f(x)$ = Output between the 0 and 1 value.

x = input to the function

e = base of natural logarithm.

When we provide the input values (data) to the function, it gives the S-curve as above.

Finding the best fit line or performance evaluation

For Linear Regression, we use the **Mean Squared Error (MSE)** cost function, which is the average of squared error occurred between the predicted values and actual values. It can be written as:

For the above linear equation, MSE can be calculated as:

$$\text{MSE} = \frac{1}{N} \sum_{i=1}^n (y_i - (a_1 x_i + a_0))^2$$

Where,

N = Total number of observation

Y_i = Actual value

$(a_1 x_i + a_0)$ = Predicted value.

Residuals: The distance between the actual value and predicted values is called **residual**. If the observed points are far from the regression line, then the residual will be high, and so cost function will high. If the scatter points are close to the regression line, then the residual will be small and hence the cost function.

t test

F test (ANOVA)

The coefficient of determination or R^2 or R-squared or (R-Sq)

This statistic is the ratio of the sum of squares of the predicted responses (corrected for the mean) to the sum of squares of the observed responses (also corrected for the mean):

$$R^2 = \text{Sum of squares (Model)} / \text{Sum of Squares (observed)}$$

Finding the best fit line or performance evaluation

Gradient Descent:

Gradient descent is used to minimize the MSE by calculating the gradient of the cost function.

A regression model uses gradient descent to update the coefficients of the line by reducing the cost function.

It is done by a random selection of values of coefficient and then iteratively update the values to reach the minimum cost function.

The process of finding the best model out of various models is called **optimization**. It can be achieved by below method:

R-squared method:

- R-squared is a statistical method that determines the goodness of fit.
- It measures the strength of the relationship between the dependent and independent variables on a scale of 0-100%.
- The high value of R-square determines the less difference between the predicted values and actual values and hence represents a good model.
- It is also called a **coefficient of determination**, or **coefficient of multiple determination** for multiple regression.
- It can be calculated from the below formula:

$$R\text{-squared} = \frac{\text{Explained variation}}{\text{Total Variation}}$$

R-squared method

Study Hours (X)	Actual Marks (Y)	Predicted Marks ($\hat{Y}$)
1	52	50
2	55	54
3	61	60
4	66	65
5	70	72

Step 1: Formula of R^2

$$R^2 = 1 - \frac{\sum(Y - \hat{Y})^2}{\sum(Y - \bar{Y})^2}$$

Where:

- Y = actual value
- $\hat{Y}$ = predicted value
- $\bar{Y}$ = mean of actual values

Step 2: Calculate Mean of Actual Values

$$\bar{Y} = \frac{52 + 55 + 61 + 66 + 70}{5}$$

$$\bar{Y} = \frac{304}{5} = 60.8$$

Y	$\hat{Y}$	$(Y - \hat{Y})$	$(Y - \hat{Y})^2$	$(Y - \bar{Y})$	$(Y - \bar{Y})^2$
52	50	2	4	-8.8	77.44
55	54	1	1	-5.8	33.64
61	60	1	1	0.2	0.04
66	65	1	1	5.2	27.04
70	72	-2	4	9.2	84.64

Sum values

$$\sum (Y - \hat{Y})^2 = 4 + 1 + 1 + 1 + 4 = 11$$

$$\sum (Y - \bar{Y})^2 = 77.44 + 33.64 + 0.04 + 27.04 + 84.64 = 222.8$$

Step 4: Calculate R^2

$$R^2 = 1 - \frac{11}{222.8}$$

$$R^2 = 1 - 0.0493$$

$$R^2 = 0.9507$$

Supervise: Classification

2. Classification

- Random Forest
- Decision Trees
- Logistic Regression
- Support vector Machines

Classification Algorithms can be further divided into the Mainly two category:

•Linear Models

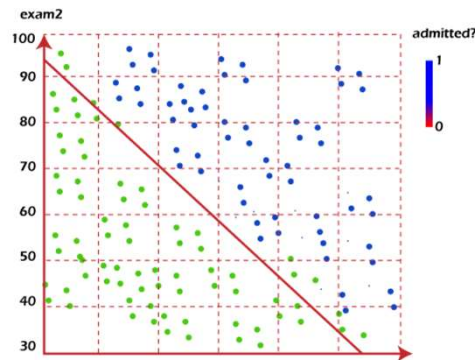
- Logistic Regression
- Support Vector Machines

•Non-linear Models

- K-Nearest Neighbours
- Kernel SVM
- Naïve Bayes
- Decision Tree Classification
- Random Forest Classification

For example, logistic regression can take two test scores for a student to predict that the student will get admission to a particular college. Because the guess is a probability, the output is a number between 0 and 1, where 1 represents absolute certainty. For the student, if the predicted probability is greater than 0.5, we estimate that they will be admitted. If the predicted probability is less than 0.5, we estimate it will be rejected.

The chart below shows the marks of past students and whether they were admitted. Logistic regression allows us to draw a line that represents the decision boundary.



Classification

Types of ML Classification Algorithms:

Classification Algorithms can be further divided into the following types:

- Logistic Regression
- K-Nearest Neighbours
- Support Vector Machines
- Kernel SVM
- Naïve Bayes
- Decision Tree Classification
- Random Forest Classification

Evaluating a Classification model:

Once our model is completed, it is necessary to evaluate its performance; either it is a Classification or Regression model. So for evaluating a Classification model, we have the following ways:

1. Log Loss or Cross-Entropy Loss:

- It is used for evaluating the performance of a classifier, whose output is a probability value between the 0 and 1.
- For a good binary Classification model, the value of log loss should be near to 0.
- The value of log loss increases if the predicted value deviates from the actual value.
- The lower log loss represents the higher accuracy of the model.
- For Binary classification, cross-entropy can be calculated as:

$$-(y \log(p) + (1-y) \log(1-p))$$

Where y= Actual output, p= predicted output.

Cross-Entropy loss

We can measure the error between two probability distributions using the cross-entropy loss function. For example, let's assume we're conducting a binary classification task (a classification task with two classes, 0 and 1).

In this instance, we must use binary cross-entropy, which is the average cross-entropy across all data samples:

$$L = -\frac{1}{N} \left[\sum_{j=1}^N [t_j \log(p_j) + (1 - t_j) \log(1 - p_j)] \right]$$

for N data points where t_i is the truth value taking a value 0 or 1 and p_i is the Softmax probability for the i^{th} data point.

Confusion Matrix also called Error Matrix

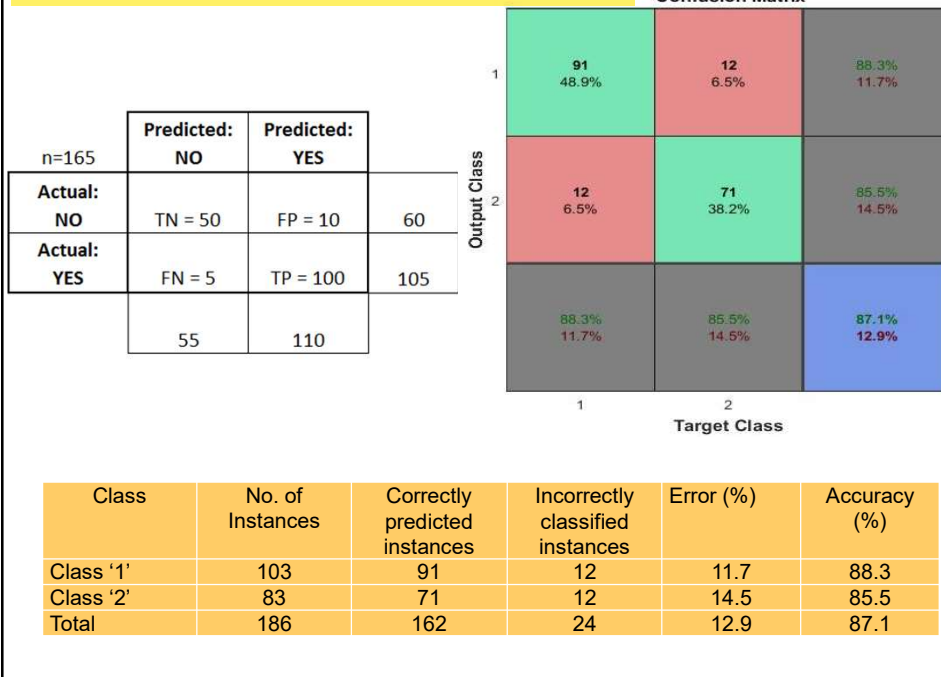
2. Confusion Matrix:

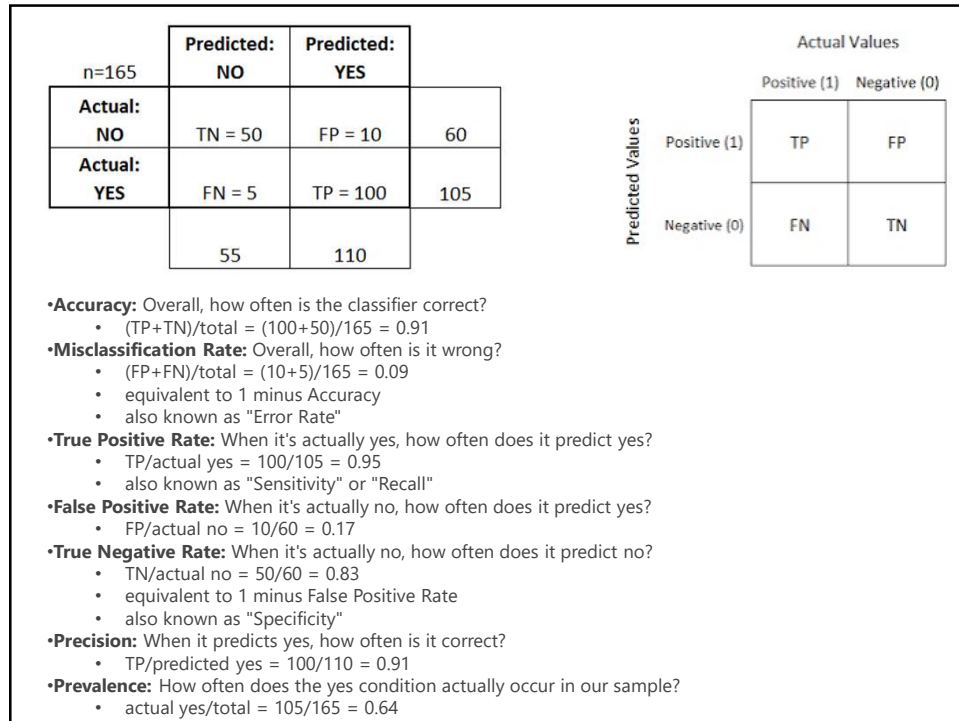
- The confusion matrix provides us a matrix/table as output and describes the performance of the model.
- It is also known as the error matrix.
- The matrix consists of predictions result in a summarized form, which has a total number of correct predictions and incorrect predictions. The matrix looks like as below table:

	Actual Positive	Actual Negative
Predicted Positive	True Positive	False Positive
Predicted Negative	False Negative	True Negative

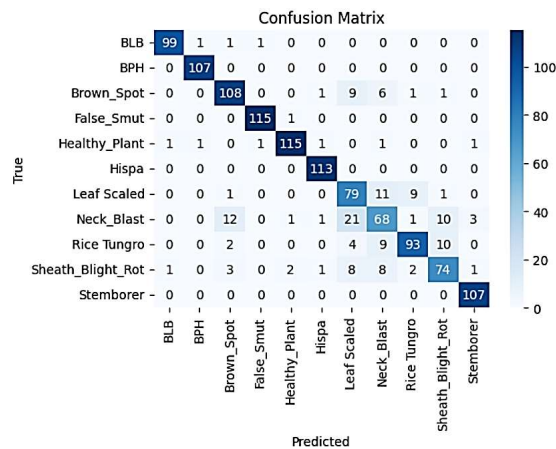
$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{Total Population}}$$

Confusion Matrix also called Error Matrix





Confusion Matrix also called Error Matrix



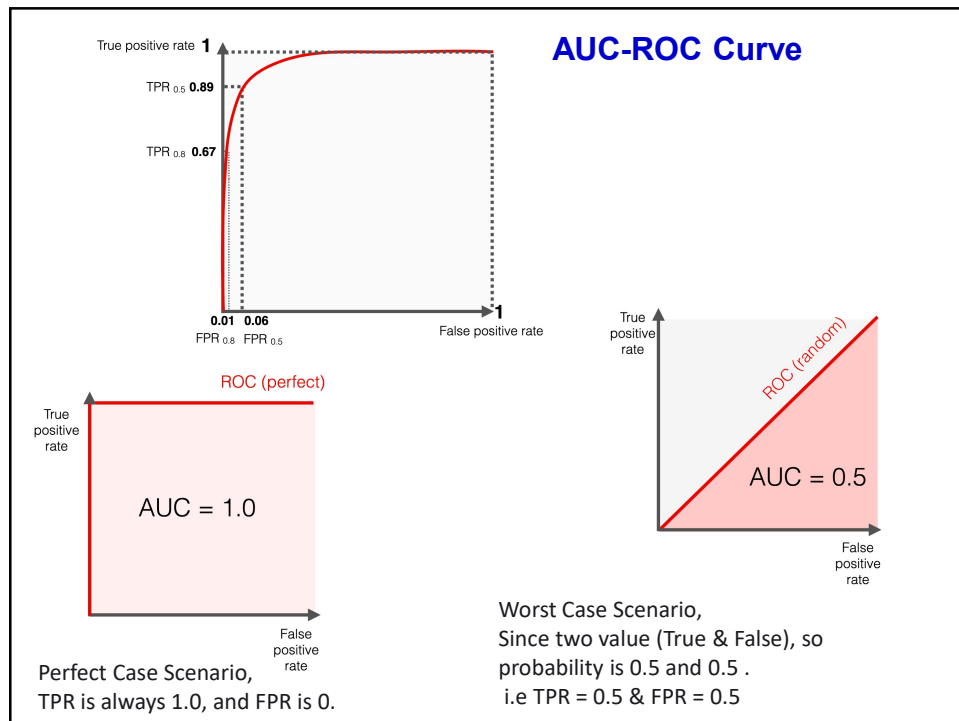
AUC-ROC Curve

ROC : Receiver Operating Characteristics Curve

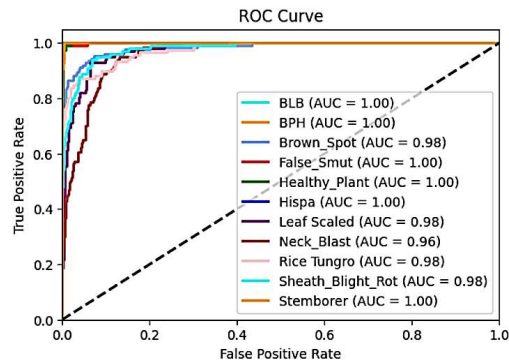
AUC : Area Under the Curve

3. AUC-ROC curve:

- ROC curve stands for **Receiver Operating Characteristics Curve** and AUC stands for **Area Under the Curve**.
- It is a graph that shows the performance of the classification model at different thresholds.
- To visualize the performance of the multi-class classification model, we use the AUC-ROC Curve.
- The ROC curve is plotted with TPR and FPR, where TPR (True Positive Rate) on Y-axis and FPR(False Positive Rate) on X-axis.



AUC-ROC Curve



Recall and F1 Score

$$\text{Recall} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Negatives}}$$

The F1 score can be interpreted as a harmonic mean of the precision and recall,

where an F1 score reaches its best value at 1 and worst score at 0.

The relative contribution of precision and recall to the F1 score are equal. The formula for the F1 score is:

$$F1 = \frac{2 * TP}{2 * TP + FP + FN}$$

where, TP is the number of true positives

FN is the number of false negatives, and

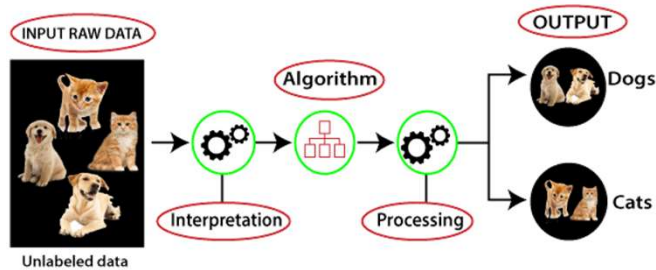
FP is the number of false positives.

F1 is by default calculated as 0.0 when there are no true positives, false negatives, or false positives.

Unsupervised Machine Learning

In unsupervised machine learning, the machine is trained using the unlabeled dataset, and the machine predicts the output without any supervision.

Working of unsupervised learning can be understood by the below diagram:

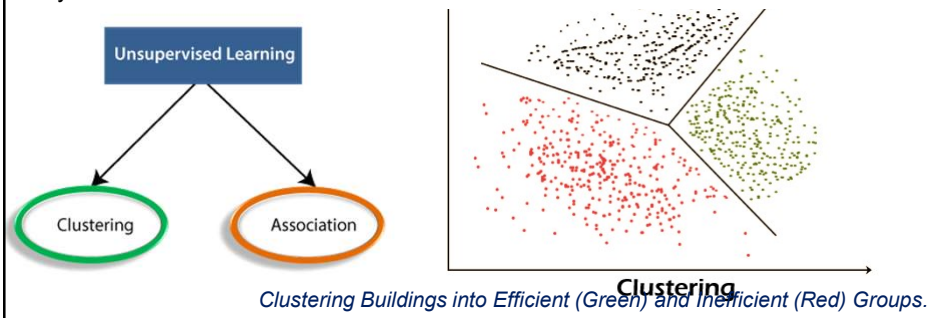


Types of Unsupervised Learning Algorithm:

The unsupervised learning algorithm can be categorized into two types of problems:

Clustering: Clustering is a method of grouping the objects into clusters such that objects with most similarities remain in a group and have less or no similarities with the objects of another group.

Association: An association rule is an unsupervised learning method which is used for finding the relationships between variables in the large database. It determines the set of items that occurs together in the dataset. Association rule makes marketing strategy more effective. Such as people who buy X item (suppose a bread) are also tend to purchase Y (Butter/Jam) item. A typical example of Association rule is Market Basket Analysis.



Types of Unsupervised Machine Learning

1. Clustering and

2. Association

1. Clustering:

Finding a structure or pattern in a collection of uncategorized data. We can also modify how many clusters your algorithm should identify.



Fig. 1

Examples:

- Finding fraudulent transactions by clustering different transaction types.

Types of Unsupervised Machine Learning

2. Association :

It allows us to establish associations amongst data objects inside large databases. This is about discovering interesting relationships between variables in large databases.

Examples:

- People that buy new home most likely to buy new furniture.

Most Used Algorithms:

- Restricted Boltzmann Machine (RBM)
- Deep Belief Network (DBN)
- Autoencoders for dimensionality reduction

Unsupervised Learning algorithms:

Below is the list of some popular unsupervised learning algorithms:

- **K-means clustering**
- **KNN (k-nearest neighbors)**
- **Hierarchical clustering**
- **Anomaly detection**
- **Neural Networks**
- **Principal Component Analysis**
- **Independent Component Analysis**
- **Apriori algorithm**
- **Singular value decomposition**

Some of the popular clustering algorithms are given below:

- **K-Means Clustering algorithm**
- **Mean-shift algorithm**
- **DBSCAN Algorithm**
- **Principal Component Analysis**
- **Independent Component Analysis**

Some popular algorithms of Association rule learning are

- **Apriori Algorithm**
- **Eclat**
- **FP-growth algorithm.**

Reinforcement Learning

Reinforcement learning works on a feedback-based process, in which an AI agent (A software component) automatically explore its surrounding by hitting & trail, taking action, learning from experiences, and improving its performance.

Agent gets rewarded for each good action and get punished for each bad action; hence the goal of reinforcement learning agent is to maximize the rewards.

Reinforcement learning is employed in different fields such as **Game theory, Operation Research, Information theory, multi-agent systems.**

A reinforcement learning problem can be formalized using **Markov Decision Process(MDP).**

Reinforcement Learning

Categories of Reinforcement Learning

Reinforcement learning is categorized mainly into two types of methods/algorithms:

Positive Reinforcement Learning: Positive reinforcement learning specifies increasing the tendency that the required behavior would occur again by adding something. It enhances the strength of the behavior of the agent and positively impacts it.

Negative Reinforcement Learning: Negative reinforcement learning works exactly opposite to the positive RL. It increases the tendency that the specific behavior would occur again by avoiding the negative condition.

Reinforcement Machine Learning Examples & Use

Most Used Algorithm Q-learning

Examples:

- Machine playing PACMAN or other similar game.

Use Cases:

- Used in Robotics for Industrial Automation
- Used to create training systems that provides custom instruction and materials according to the requirements of students

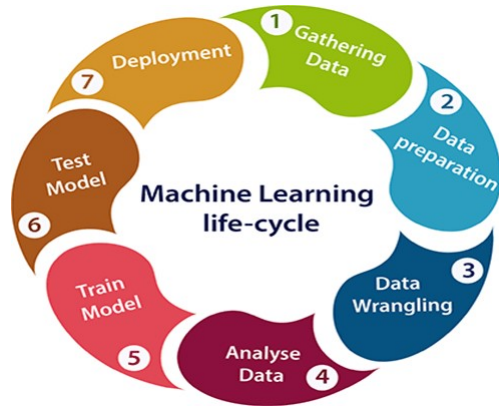
How does a machine learning system work?

Machine learning life cycle is a cyclic process to build an efficient machine learning project.

The main purpose of the life cycle is to find a solution to the problem or project.

Machine learning life cycle involves seven major steps, which are given below:

1. Gathering Data
2. Data preparation
3. Data Wrangling
4. Analyse Data
5. Train the model
6. Test the model
7. Deployment



How does a machine learning system work?

1. Gathering Data:

This step includes the below tasks:

- Identify various data sources
- Collect data
- Integrate the data obtained from different sources

By performing the above task, we get a coherent set of data, also called as a **dataset**. It will be used in further steps.

2. Data preparation

In this step, first, we put all data together, and then randomize the ordering of data.

- **Data exploration:**
It is used to understand the nature of data that we have to work with. We need to understand the characteristics, format, and quality of data.
- **Data pre-processing:**
Now the next step is preprocessing of data for its analysis.

How does a machine learning system work?

3. Data Wrangling

It is the process of **cleaning the data**, selecting the variable to use, and **transforming the data in a proper format** to make it more suitable for analysis in the next step.

It is not necessary that data we have collected is always of our use as some of the data may not be useful. In collected data may have various issues, including:

- i. **Missing Values**
- ii. **Duplicate data**
- iii. **Invalid data**

It is mandatory to detect and remove the above issues because it can negatively affect the quality of the outcome.

How does a machine learning system work?

4. Data Analysis

This step involves:

- **Selection of analytical techniques**
- **Building models**
- **Review the result**

where we select the machine learning techniques such as **Classification, Regression, Cluster analysis, Association**, etc. then build the model using prepared data, and evaluate the model.

5. Train Model

In this step we **train our model to improve its performance for better outcome of the problem.**

6. Test Model

Once our machine learning model has been trained on a given dataset, then we test the model.

In this step, **we check for the accuracy of our model by providing a test dataset to it.**

7. Deployment

The last step of machine learning life cycle is deployment, where we deploy the model in the real-world system.

If the above-prepared model is producing an accurate result as per our requirement with acceptable speed, then we deploy the model in the real system. But before deploying the project, we will check whether it is improving its performance using available data or not. The deployment phase is similar to making the final report for a project.

Why do we need Data Preprocessing?

A real-world data generally contains noises, missing values, and maybe in an unusable format which cannot be directly used for machine learning models. Data preprocessing is required tasks for cleaning the data and making it suitable for a machine learning model which also increases the accuracy and efficiency of a machine learning model.

It involves below steps:

1. **Getting the dataset**
2. **Importing libraries**
3. **Importing datasets**
4. **Finding Missing Data**
5. **Encoding Categorical Data**
6. **Splitting dataset into training and test set**
7. **Feature scaling**

4) Handling Missing data:

If our dataset contains some missing data, then it may create a huge problem for our machine learning model.

There are mainly two ways to handle missing data, which are:

By deleting the particular row: The first way is used to commonly deal with null values. In this way, we just delete the specific row or column which consists of null values. But this way is not so efficient and removing data may lead to loss of information which will not give the accurate output.

By calculating the mean: In this way, we will calculate the mean of that column or row which contains any missing value and will put it on the place of missing value. This strategy is useful for the features which have numeric data such as age, salary, year, etc. Here, we will use this approach.

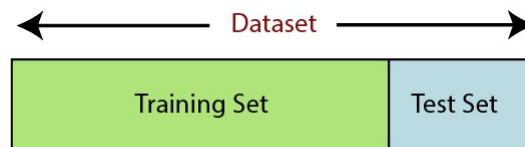
5) Encoding Categorical data:

Categorical data is data which has some categories such as, in our dataset; there are two categorical variable, **Country**, and **Purchased**.

Since machine learning model completely works on mathematics and numbers, but if our dataset would have a categorical variable, then it may create trouble while building the model. So it is necessary to encode these categorical variables into numbers.

6) Splitting the Dataset into the Training set and Test set

If we train our model very well and its training accuracy is also very high, but we provide a new dataset to it, then it will decrease the performance. So we always try to make a machine learning model which performs well with the training set and also with the test dataset. Here, we can define these datasets as:



Training Set: A subset of dataset to train the machine learning model, and we already know the output.

Test set: A subset of dataset to test the machine learning model, and by using the test set, model predicts the output.

7) Feature Scaling

Feature scaling is the final step of data preprocessing in machine learning. It is a technique to standardize the independent variables of the dataset in a specific range. In feature scaling, we put our variables in the same range and in the same scale so that no any variable dominate the other variable.

Overfitting in Machine Learning

In the real world, the dataset present will never be clean and perfect. It means each dataset contains impurities, noisy data, outliers, missing data, or imbalanced data. Due to these impurities, different problems occur that affect the accuracy and the performance of the model. One of such problems is Overfitting in Machine Learning. *Overfitting is a problem that a model can exhibit.*

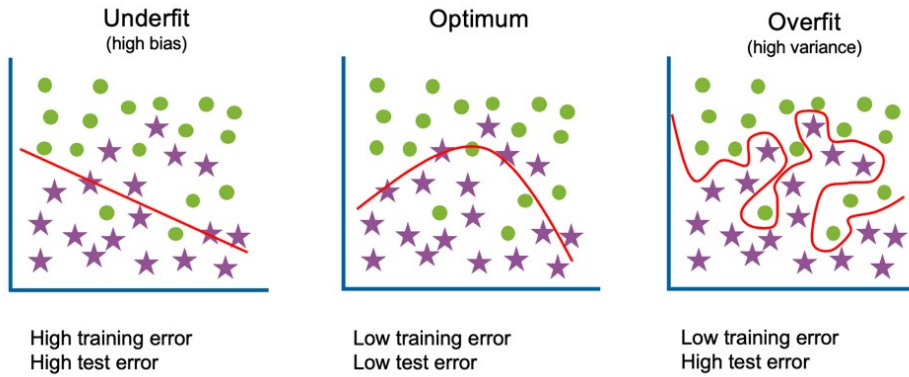
Noise: Noise is meaningless or irrelevant data present in the dataset. It affects the performance of the model if it is not removed.

Bias: Bias is a prediction error that is introduced in the model due to oversimplifying the machine learning algorithms. Or it is the difference between the predicted values and the actual values.

Variance: If the machine learning model performs well with the training dataset, but does not perform well with the test dataset, then variance occurs.

Generalization: It shows how well a model is trained to predict unseen data.

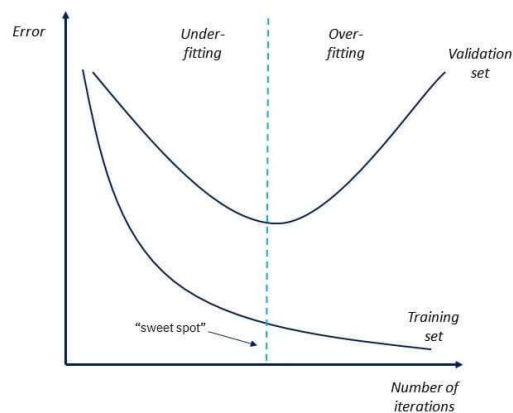
What is Overfitting?



How to detect Overfitting?

Now, if the model performs well with the training dataset but not with the test dataset, then it is likely to have an overfitting issue.

For example, if the model shows 85% accuracy with training data and 50% accuracy with the test dataset, it means the model is not performing well.



Ways to prevent the Overfitting

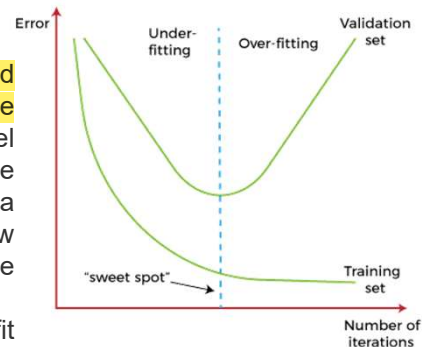
Below are several ways that can be used to prevent overfitting:

1. **Early Stopping**
2. **Train with more data**
3. **Feature Selection**
4. **Cross-Validation**
5. **Data Augmentation**
6. **Regularization**

Early Stopping

In this technique, the training is paused before the model starts learning the noise within the model. While training the model iteratively, measure the performance of the model after each iteration. Continue up to a certain number of iterations until a new iteration improves the performance of the model.

After that point, the model begins to overfit the training data; hence we need to stop the process before the learner passes that



Train with More data

When a model is fed with more training data, it will be unable to overfit all the samples of data and forced to generalize well.

But in some cases, the additional data may add more noise to the model; hence we need to be sure that data is clean and free from in-consistencies before feeding it to the model.

Feature Selection

Sometimes some of the features are redundant or less important for the prediction, and for this feature selection process is applied. In the feature selection process, we identify the most important features within training data, and other features are removed. Further, this process helps to simplify the model and reduces noise from the data.

Some algorithms have the auto-feature selection, and if not, then we can manually perform this process.

Cross-Validation

Cross-validation is one of the powerful techniques to prevent overfitting.

In the general k-fold cross-validation technique, we divided the dataset into k-equal-sized subsets of data; these subsets are known as folds.

Data Augmentation

Data Augmentation is a data analysis technique, which is an alternative to adding more data to prevent overfitting. In this technique, instead of adding more training data, slightly modified copies of already existing data are added to the dataset.

The data augmentation technique makes it possible to appear data sample slightly different every time it is processed by the model. Hence each data set appears unique to the model and prevents overfitting.

How to choose the best model?

In the above section, we have discussed different machine learning models and algorithms. But one most confusing question that may arise to any beginner that "which model should I choose?". So, the answer is that it depends mainly on the business requirement or project requirement. Apart from this, it also depends on associated attributes, the volume of the available dataset, the number of features, complexity, etc. However, in practice, it is recommended that we always start with the simplest model that can be applied to the particular problem and then gradually enhance the complexity & test the accuracy with the help of parameter tuning and cross-validation.

Are machine learning models, and algorithms are the same?

The answer to this question is No, and the machine learning model is not the same as an algorithm.

An ML algorithm is like a procedure or method that runs on data to discover patterns from it and generate the model.

A machine learning model is like a computer program that generates output or makes predictions. More specifically, when we train an algorithm with data, it becomes a model.

1. **Machine Learning Model = Model Data + Prediction Algorithm**

How does a machine learning system work?

4. What is the difference between AI, Machine Learning, and Deep Learning

Answer:

AI is the broader concept of machines performing tasks intelligently.

Machine Learning (ML) is a subset of AI where machines learn from data.

Deep Learning (DL) is a subset of ML using neural networks with many layers.

7. What are some common algorithms used in AI labs?

Answer:

- c Search Algorithms (DFS, BFS, A*)
- Classification (KNN, SVM, Decision Tree)
- Clustering (K-Means)
- Regression (Linear, Polynomial)
- Reinforcement Learning (Q-Learning)
- Neural Networks

